

A Project Report
On
Design Of MIMO Antennas

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**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI
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Hyderabad Campus**

Certificate

This is to certify that the project report entitled “**Design Of MIMO Antennas**” submitted by Debanshu Ghosh (ID No. 2023A3PS0332H) in partial fulfillment of the requirements of the course EEE F76, Design Project Course, embodies the work done by him under my supervision and guidance.

Date: 13/10/2025

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ABSTRACT

This project focuses on the design, simulation, and analysis of Multiple-Input Multiple-Output (MIMO) antennas for modern wireless communication systems. MIMO technology enhances system capacity, reliability, and spectral efficiency by utilizing multiple transmitting and receiving antennas.

In this work, microstrip patch antennas are designed and optimized using full-wave electromagnetic simulation tools to *operate at 10 GHz*, a frequency relevant to emerging high-speed communication bands. Various antenna parameters such as *return loss (S_{11})*, *mutual coupling (S_{21})*, *radiation pattern and gain* are analyzed to evaluate performance. Design techniques to improve isolation between antenna elements—such as optimizing element spacing and substrate properties—are explored to achieve low coupling and stable radiation characteristics.

The results demonstrate that with proper impedance matching and spatial configuration, compact MIMO antenna systems can achieve efficient operation with reduced mutual coupling, making them suitable for next-generation wireless and radar applications.

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Introduction

The increasing demand for high-speed and reliable wireless communication has led to the evolution of advanced antenna technologies. Among these, Multiple-Input Multiple-Output (MIMO) systems have become a cornerstone of modern wireless standards due to their ability to improve data throughput, link reliability, and spectral efficiency without additional bandwidth or power requirements.

This project focuses on the **design and analysis of microstrip patch antennas for MIMO applications** operating around **10 GHz**. The work involves understanding antenna theory, studying existing research and simulating patch configurations to evaluate parameters such as **return loss (S_{11})**, **mutual coupling (S_{21})** and **radiation characteristics**. The objective is to achieve efficient radiation, low reflection and good isolation between antenna elements—key factors that determine the performance of MIMO systems in modern communication environments.

Literature Survey

This report reviews key principles in antenna theory, moving from foundational single-element designs to advanced multi-antenna systems, particularly focusing on printed compact solutions. The theoretical groundwork involves the design and analysis of microstrip patch antennas and arrays for applications such as GPS and satellite communication, emphasizing the trade-off between miniaturization and performance parameters like resonant frequency and impedance matching.

1. Microstrip Patch Antenna Design Procedure and Results

The design process for a compact rectangular microstrip patch antenna typically involves calculating dimensions based on the desired resonant frequency, substrate dielectric constant, and substrate height.

- **Design Parameters:** The length and width of the patch are initially calculated using approximate transmission line model formulas (e.g., Equations (5.1) and (5.4) in Alsager, 2011).
- **Substrate Effect:** The use of a higher dielectric constant material drastically **reduces the physical dimensions** of the patch, enabling miniaturization, which is critical for compact devices like GPS receivers or satellite TV terminals.
- **Feeding and Matching:** A key step is determining the optimal feed point position () to achieve the desired **input impedance** for maximum power transfer. This often requires iterative adjustment from initial calculation, as highlighted in the GPS antenna example.
- **Array Application:** Combining multiple MPAs into an array (e.g., 2-element or 4-element) can enhance performance, notably **increasing received power and directivity**, which is desirable for applications like satellite communication. The challenge then shifts to designing the corporate feed network and matching lines to maintain system impedance.

2. In-Depth Analysis of Critical MIMO Performance Metrics

The reliable characterization of MIMO antenna systems necessitates specific metrics that account for the interaction between multiple closely spaced elements and the surrounding environment.

A. Correlation Coefficient (CC) or Envelope Correlation Coefficient (ECC)

The ECC () quantifies the statistical independence of the radiation channels, which is paramount for achieving the theoretical channel capacity gains of MIMO. A value below **0.3** is often considered acceptable for 4G wireless systems.

- **Calculation Method:** The accurate method requires integrating the **three-dimensional (3D) radiated field patterns** () of the antenna elements (Equation (2) in Sharawi, 2013) to correctly capture the pattern coupling.
- **Common Misuse:** A frequent error is calculating ECC solely from **S-parameters** (e.g., Equation (4) in Sharawi, 2017). This method is **only valid for lossless antennas** in an isotropic environment and fails for practical, lossy printed antennas, incorrectly suggesting good correlation despite significant field overlap.
- **Isolation vs. Correlation:** **Port Isolation** (high) aims to prevent power leakage between ports, improving efficiency. **Field Correlation** aims to decouple the spatial fields, improving capacity. High isolation does *not* guarantee low correlation unless the antenna patterns are spatially isolated (e.g., orthogonal polarization or tilted beams).

B. Total Active Reflection Coefficient (TARC)

TARC () is a better measure of the MIMO system's total **impedance bandwidth and efficiency** than single-port reflection coefficients (). It is the ratio of the square root of total reflected power to the square root of total incident power across all ports.

- TARC inherently accounts for the effects of mutual coupling and the phase differences between input signals, making it a comprehensive measure of the system's operational bandwidth for various excitation conditions.

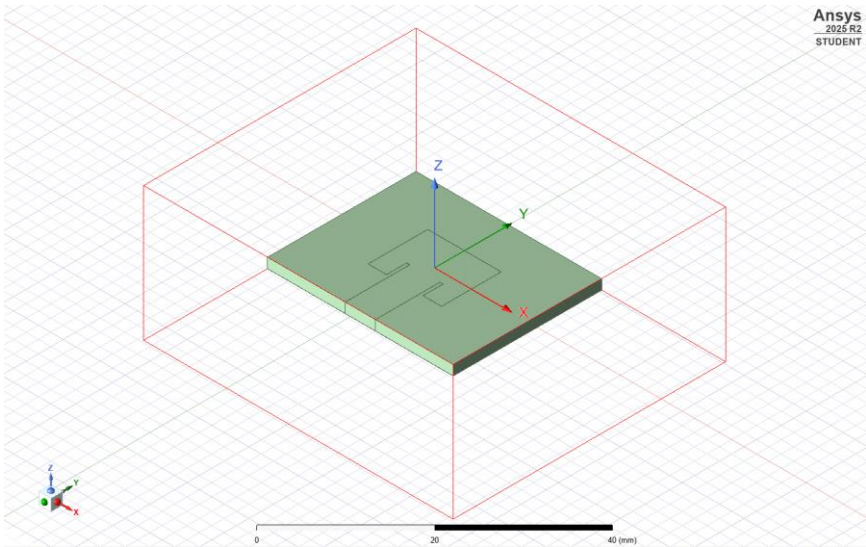
3. Comparative Literature Survey: Microstrip Antennas and MIMO Systems

This section synthesizes the fundamental antenna principles discussed in the thesis "Design and Analysis of Microstrip Patch Antenna Arrays" with the advanced MIMO (Multiple-Input Multiple-Output) concepts and challenges highlighted in the two articles by Sharawi.

Feature	Microstrip Patch Antennas (Alsager, 2011)	MIMO Antenna Systems (Sharawi, 2013 & 2017)
Primary Goal	Single Element/Array Design: Achieve resonance, directional patterns, high gain, and impedance matching.	Multi-element Integration: Maximize Channel Capacity by utilizing spatial diversity and increasing the number of antennas.
Key Antenna Type	Rectangular and circular patch designs.	Printed monopoles, PIFAs, F-shaped, and dielectric resonator antenna (DRA) elements.
Critical Metric	Return Loss (S11): Measures power delivered to the antenna, ensuring good impedance matching to the source.	Envelope Correlation Coefficient (or ECC): Measures channel isolation/independence; directly impacts channel capacity. Total Active Reflection Coefficient (TARC): Assesses system bandwidth and efficiency under multi-port excitation.
Isolation/Coupling	Focus is on minimizing reflected power at the feed point. Mutual effects in arrays (less than spacing) complicate calculations.	Crucial Challenge: High mutual coupling, especially at lower frequency bands () in closely packed elements, degrades performance. Solutions include DGS, neutralization lines, and metamaterials.
Design Trade-offs	Size Reduction: High dielectric constant reduces size but may affect efficiency and bandwidth.	Miniaturization vs. Performance: Antenna miniaturization negatively affects efficiency and operating bandwidth.

Design and Simulated Performance of the Microstrip Patch Antenna

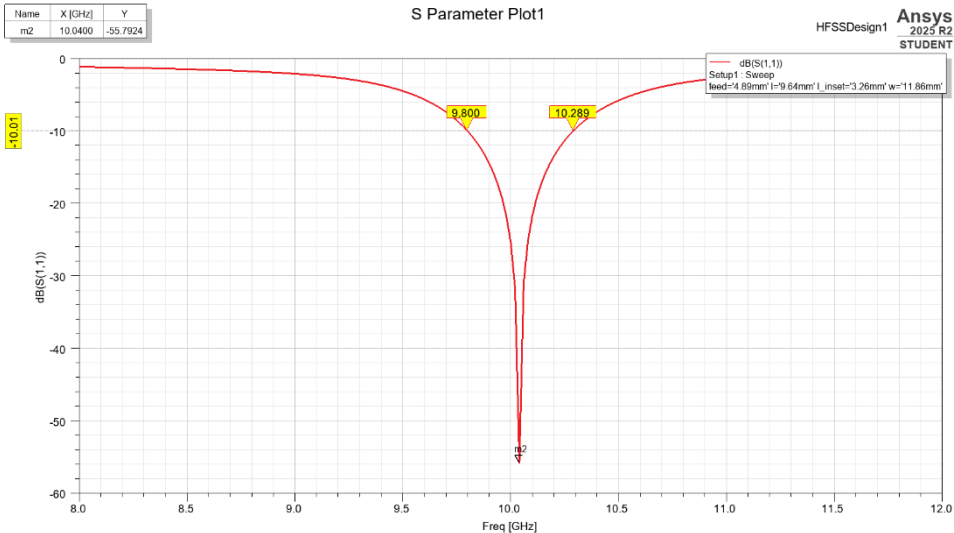
This design represents a single-element microstrip patch antenna specifically optimized for operation at 10GHz within the high-frequency X-band (8–12 GHz) spectrum, leveraging the advantages of planar, low-profile technology. The core radiating element is a modified rectangular patch with precise physical dimensions: **a length of 9.64mm** and a **width of 11.86mm**. The antenna is situated on a dielectric substrate with a **thickness of 1.588 mm**. U-shaped slots are etched into the patch surface; these are employed to fine-tune the element's complex impedance and precise resonant frequency. Power is coupled to the patch using a direct inset feed mechanism, where the feed line taps the radiating patch at an optimized **inset depth of 3.26mm** , a crucial parameter for achieving a precise $50\ \Omega$ impedance match.



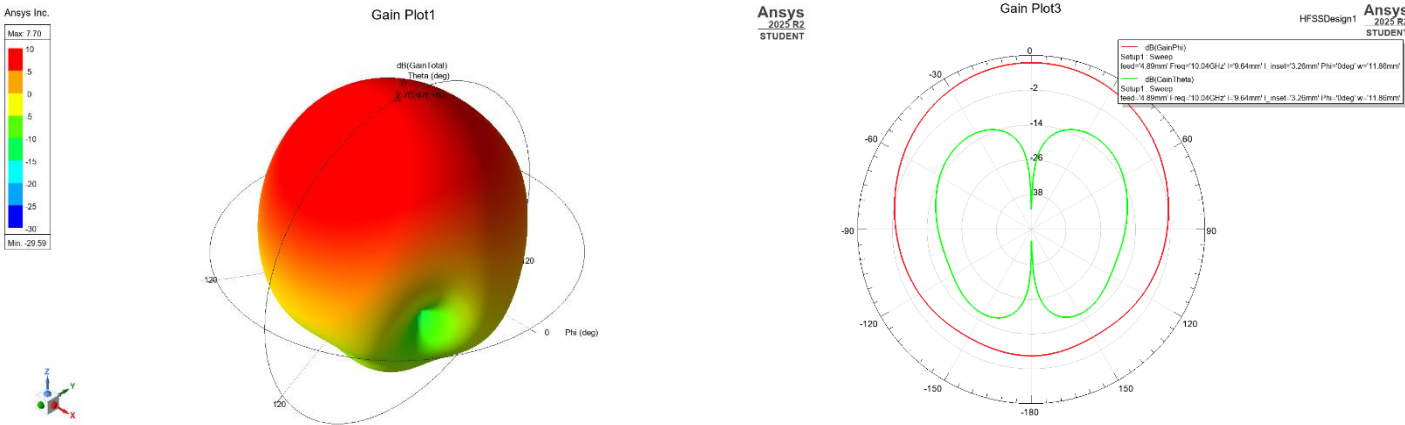
Name	Value	Unit
a	30	mm
b	24	mm
h	1.588	mm
t	0.017	mm
w	11.86	mm
l	9.64	mm
feed	4.89	mm
l_inset	3.26	mm

- Impedance Matching and S-Parameter Performance**

The simulation results for the reflection coefficient (S_{11} or Return Loss) demonstrate an extraordinary level of numerical precision in impedance matching, characteristic of a high-Quality factor (Q) design. The antenna is tuned to resonate at a precise **frequency of 10.040GHz** , where the return loss achieves a remarkable **minimum value of -55.792dB** . This extreme match signifies that a negligible fraction of the incident power is reflected back to the source. The system provides a usable operational bandwidth, defined by the standard threshold of $S_{11} < -10\text{dB}$, spanning from **9.800 GHz to 10.289 GHz** . This results in an operational bandwidth of approximately 489 Mhz and **Fractional Bandwidth of 4.8%**

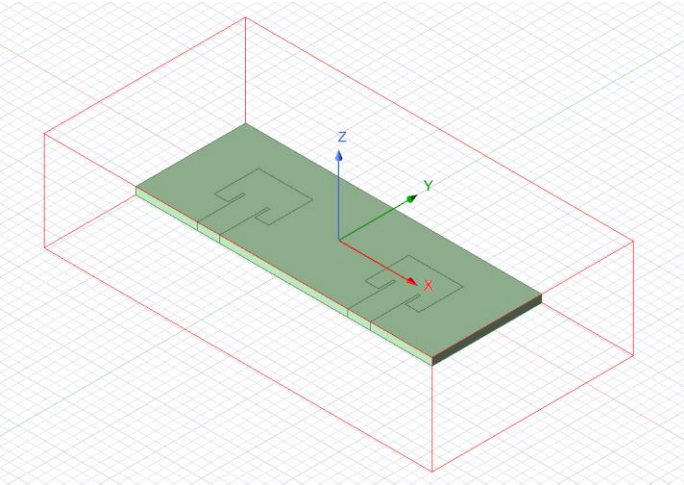


The far-field simulation confirms that the antenna operates as a highly effective radiator with characteristics suitable for directional applications within the X-band. The element exhibits a classic **broadside radiation pattern**, with the main beam directed perpendicularly away from the antenna's planar surface. This pattern is essential for high-gain applications such as fixed radar or satellite communication. The antenna achieves a **high peak realized gain of 7.70dBi**, demonstrating efficient conversion of input power into radiated energy. Analysis of the two-dimensional polar plots shows the angular distribution of the radiated power, confirming the dominance of the desired co-polar component and affirming the design's success in generating a focused, linearly polarized beam with high directivity.



Design and Simulated Performance of the X-Band Dual-Element Microstrip Array

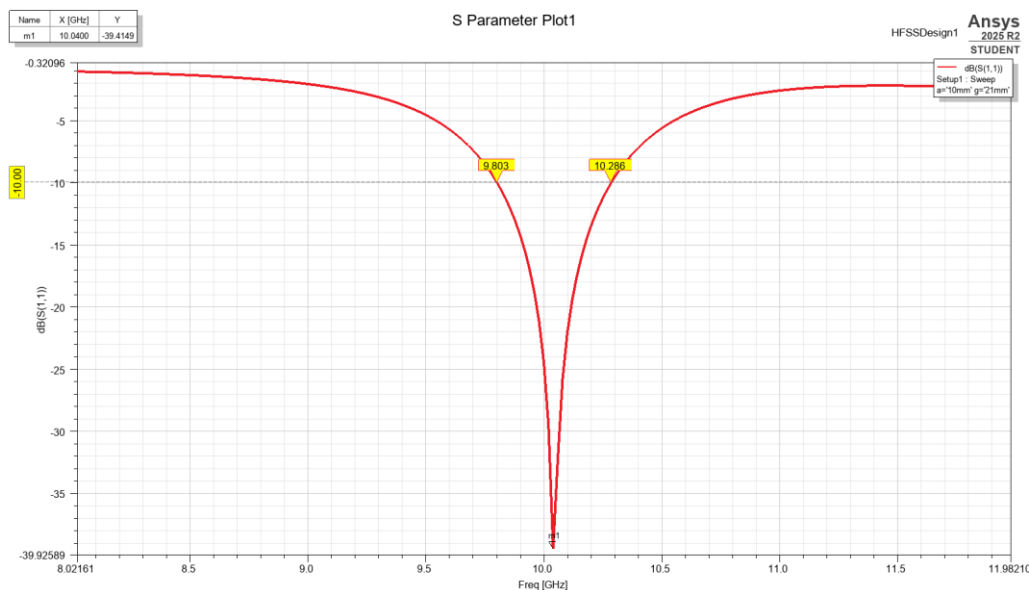
This focuses on the characteristics of a dual-element microstrip patch antenna (MPA) array designed for operation in the X-band (**approximately 10 GHz**). The array configuration consists of two identical, modified rectangular patch elements arranged symmetrically on a single dielectric substrate. The patches feature internal U-shaped slots, a circuit feature typically used to precisely adjust the resonant frequency and optimize the input impedance, particularly crucial for high-frequency tuning. The antenna elements are coupled to the feed network via an inset feed, optimized for a **50 Ω interface** and are printed on a substrate with a **thickness of 1.588mm**.



	Name	Value	Unit
	a	10	mm
	b	24	mm
	h	1.588	mm
	t	0.017	mm
	w	11.86	mm
	l	9.64	mm
	feed	4.89	mm
	l_inset	3.26	mm
	g	21	mm

• Impedance Matching Performance (S_{11})

The impedance performance of the array elements is characterized by the reflection coefficient (S_{11}), which measures the efficiency of power delivery to the antenna. The simulated results show the dual-element array with **inset depth 3.26mm, patch length 9.64mm and patch gap 21mm** is **resonant at 10.040 GHz**. At this precise operating frequency, the **return loss is measured at -39.415 dB**. This level of matching is highly effective, signifying that less than of the incident power is reflected back toward the source, ensuring high power transfer efficiency. The bandwidth of operation extends across a frequency range of approximately **460 MHz (from 9.803 GHz to 10.286 GHz)**.



• Inter-Port Isolation (S_{21}) Performance

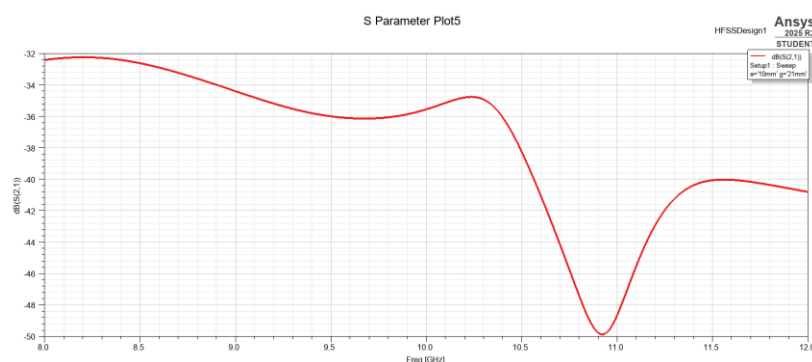
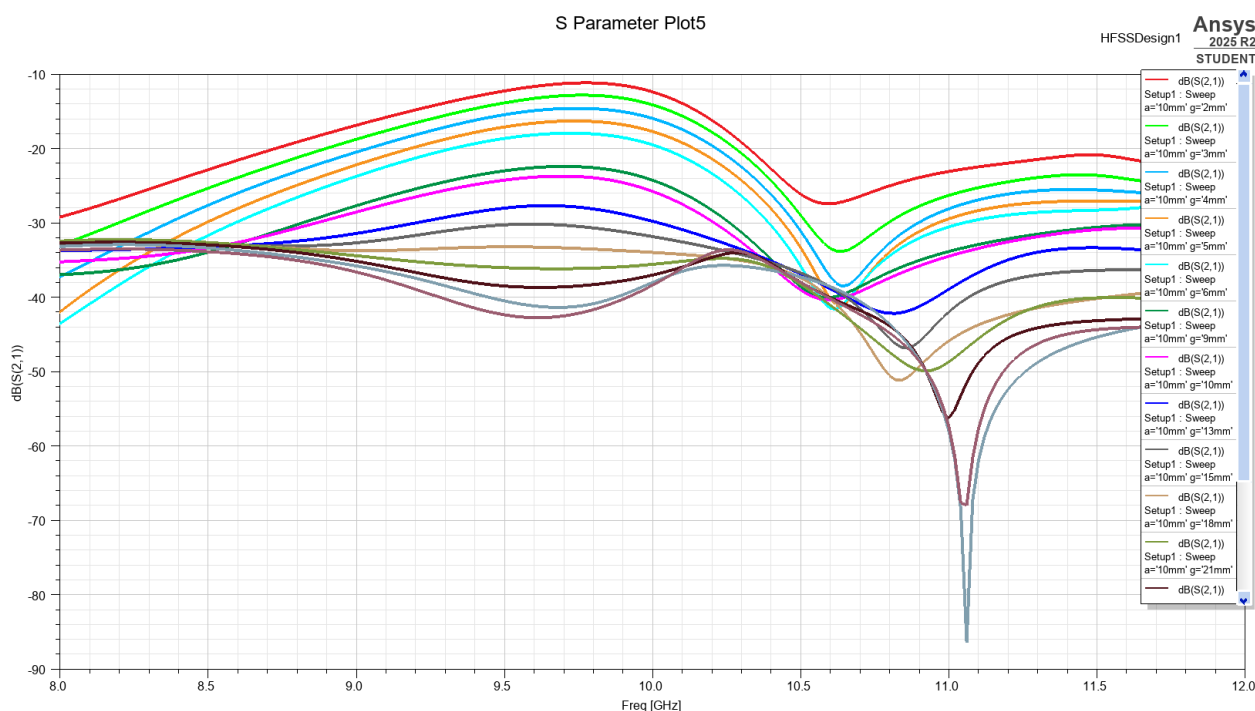
Variation of S_{21} with Gap (g)

1. Observation from the Plot

As g increases from 2 mm to 27 mm, the S_{21} curve shifts downward, meaning less coupling (more negative S_{21} values). At small gaps (2–5 mm), the coupling is strong — S_{21} is around –20 to –30 dB. As the gap increases to around 10–13 mm, the dip deepens significantly (–50 to –70 dB). Beyond 18 mm, the curves saturate — isolation improves slightly but the resonant dip remains near 10.8–11 GHz.

2. Physical Explanation

- **Coupling** occurs due to mutual electromagnetic interaction between the two patch antennas — mainly **surface waves** and **fringing fields**.
- When the gap g is **small**, the patches are close enough for strong near-field interaction → more energy leaks from one patch to the other → **higher coupling** (S_{21} less negative).
- Increasing g weakens this interaction → less mutual coupling → **lower S_{21} (more negative values)**.



Conclusion

The X-band dual-element microstrip array successfully demonstrates exceptional numerical performance in its port characteristics at the resonant frequency of **10.040 GHz**. The individual element tuning yields highly efficient power transfer, characterized by an excellent **return loss (S11) of approximately -39.415 dB** and a functional bandwidth of 460 MHz centered near 10 GHz. A key outcome of the simulation is the remarkable **inter-port isolation (S21), which reaches an extreme minimum of -49 dB** with **inset depth 3.26mm, patch length 9.64mm and patch gap 21mm**. This substantial disparity, where the coupling is significantly lower than the input match, confirms negligible energy leakage between the adjacent ports. However, this result requires critical context: established literature emphasizes that ultra-high isolation does not automatically translate to a high-performance Multiple-Input, Multiple-Output (MIMO) system; instead, system viability depends on a low Envelope Correlation Coefficient (ECC), which quantifies the statistical independence of the radiation patterns.

References

1. **Antenna theory: analysis and design** / Constantine A. Balanis, third edition, Hoboken, NJ: Wiley, 2005, ISBN 047166782X
2. **Sharawi, M. S.**, "Current Misuses and Future Prospects for Printed Multiple-Input, Multiple-Output Antenna Systems," *IEEE Antennas and Propagation Magazine*, vol. 59, no. 2, pp. 162-170, April 2017.
3. **Sharawi, M. S.**, "Printed Multi-Band MIMO Antenna Systems and Their Performance Metrics," *IEEE Antennas and Propagation Magazine*, vol. 55, no. 5, pp. 218-232, October 2013.
4. Alsager, A. F., "Design and Analysis of Microstrip Patch Antenna Arrays," Master Thesis No. 1/2011, School of Engineering, Höskolan i Borås, Sweden, January 2011.